

Hashing Concepts & Hash Functions -- Data Structures

Definition: Hashing is a technique that transforms a search key K into a table index h(K) using a Hash Function, enabling average constant time O(1) insertion, deletion, and lookup operations in a Hash Table.

1. Detailed Technical Explanation

Search Key (e.g., 108) ---> [Hash Function $h(k) = k \% 10$] ---> Index 8 in Hash Table

1. The Load Factor (**lpha**)

The load factor represents the density of items stored in a hash table of size m with n keys:

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* In Open Addressing, lpha ≤ 1. In Separate Chaining, lpha can exceed 1.

2. Common Hash Functions

1. Division Method (Modulo Arithmetic)

$h(k) = k \bmod m$

* Rule: m should be a Prime Number not close to a power of 2 or 10 to minimize clustering.

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2. Mid-Square Method

1. Square the key: k^2 .

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3. Folding Method

Divide the key digits into equal parts of size r, and sum them together (ignoring overflow carry).

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4. Multiplication Method

$h(k) = \text{floor}(m * (k * A \bmod 1))$ where $0 < A < 1$ (Knuth recommends $A = 0.6180339887$)

* *Advantage:* The choice of table size m is not critical (can be a power of 2).

3. Properties of a Good Hash Function

1. Uniform Distribution: Distributes keys evenly across all table slots to minimize collisions.

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4. Quick Recall Flow

Key -> Hash Function $h(k)$ -> Index $[0..m-1]$ | Functions: Division ($k \% m$), Mid-Square (middle of k^2), Folding (split & sum)